

**BEYOND A SINGULAR DIMENSION: A MULTIDIMENSIONAL TRUST
FRAMEWORK FOR UNDERSTANDING USER INTENTION AND USAGE OF LARGE
LANGUAGE MODELS (LLMs)**

By

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In Partial Fulfillment of the Requirements for the Degree of Bachelor of Science in Industrial
Engineering

Department of Industrial and Systems Engineering
Gokongwei College of Engineering
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Taft Avenue, Manila

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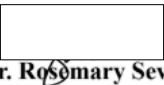
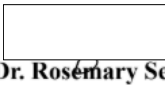
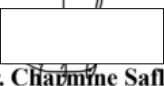
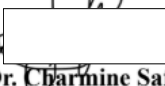
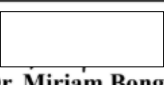
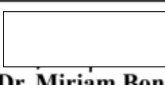
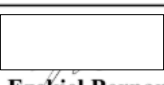
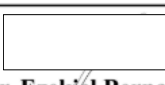
INDUSTRIAL AND SYSTEMS ENGINEERING DEPARTMENT
Gokongwei College of Engineering
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TITLE OF THE STUDY: “Beyond a Singular Dimension: A Multidimensional Trust Framework for Understanding User Intention and Usage of Large Language Models (LLMs)”

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In partial fulfillment of the requirements for the degree Bachelor of Science in Industrial Engineering has been examined for **ORAL EXAMINATION**.

PANEL OF EXAMINERS

<i>Approval Signature for Topic Proposal</i>	<i>Date</i>	<i>Approval Signature for Final Paper</i>	<i>Date</i>
 Dr. Rosemary Seva <i>Panel Chairperson</i>	12/06/2025	 Dr. Rosemary Seva <i>Panel Chairperson</i>	12/06/2025
 Dr. Charmine Safflor <i>Panel Member</i>	12/05/2025	 Dr. Charmine Safflor <i>Panel Member</i>	12/05/2025
 Dr. Miriam Bongo <i>Panel Member</i>	12/07/2025	 Dr. Miriam Bongo <i>Panel Member</i>	12/07/2025
 Dr. Ezekiel Bernardo <i>Adviser</i>	12/04/25	 Dr. Ezekiel Bernardo <i>Adviser</i>	12/04/25

Accepted and approved in partial fulfillment of the requirements for the degree of Bachelor of Science in Industrial Engineering by the Thesis Panel following the oral examination, with a passing grade.

COLLEGE AND DEPARTMENT ACCEPTANCE

<i>Name</i>	<i>Date</i>
 Dr. Richard Li <i>Department Chair, Industrial and Systems Engineering Department</i>	12/08/2025
 Dr. Kathleen B. Aviso <i>College Dean, Gokongwei College of Engineering</i>	12.08.2025

TURNITIN REPORT

THSIE4B_Beyond a Singular Dimension_ A Multidimensional Trust Framework for Understanding User Intention and Usage of Large Language Models (LLMs)

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ABSTRACT

The adoption of Large Language Models (LLMs) necessitates an understanding of how trust mechanisms influence user behavior; however, existing research employs unidimensional constructs that are inadequate for capturing the complexity of engagement. This study developed and validated a multidimensional framework disaggregating trust into cognitive (CT), Emotional (ET), Situational (ST), Dispositional (DT), History-Based (HBT), and Institutional (IT) dimensions to predict Intention to Use (IU) and Continued Engagement (CE) with LLMs. Phase 1 employed a 30-day longitudinal design with participants engaging daily with LLMs. Covariance-Based Structural Equation Modeling demonstrated that all six trust dimensions significantly predicted IU, while ET, DT, HBT, and IT significantly predicted CE. IU significantly predicted CE and partially mediated the relationship between DT and CE. Template Thematic analysis was conducted to identify design mechanisms targeting specific trust dimensions. Phase 2 employed a design validation study where validated trust dimensions from Phase 1 were translated into UI design recommendations through a triangulated design process and were evaluated for their effectiveness via a controlled experiment and tested using Welch's t-test. Phase 2 successfully targeted five trust dimensions (CT, ET, ST, HBT, and IT), IU, and CE. This research established trust as a multidimensional, dynamically calibrating construct, advancing human-AI interaction theory while providing design guidelines for LLM interfaces that support appropriate trust calibration across adoption and sustained engagement phases.

Keywords: Large Language Models, multidimensional trust, human-AI interaction, structural equation modeling, trust calibration, technology adoption

ABSTRAK

Ang paggamit ng *Large Language Models* (LLMs) ay nangangailangan ng mas malalim na pag-unawa kung paano nakakaapekto ang mga dimension ng tiwala sa pag-uugali ng mga gumagamit. Subalit, ang umiiral na mga pananaliksik ay gumagamit ng *unidimensional* na balangkas na hindi sapat upang makuha ang kompleksidad ng tiwala sa paggamit ng LLMs. Ang pag-aaral na ito ay bumubuo at nagpatunay ng isang *multidimensional* na balangkas na naghihiwalay sa tiwala sa anim na dimensyon: *Cognitive* (CT), *Emotional* (ET), *Situational* (ST), *Dispositional* (DT), *History-Based* (HBT), at *Institutional* (IT), upang maunawaan ang *Intention to Use* (IU) at *Continued Engagement* (CE) ng mga kalahok sa paggamit ng LLMs. Sa unang yugto ng pag-aaral na ito, ginamit ang 30-araw na *longitudinal* na disenyo kung saan araw-araw na nakikipag-ugnayan ang mga kalahok sa kanilang napiling plataporma ng LLMs. Ipinakita ng *Covariance-Based Structural Equation Modeling* na lahat ng anim na dimensyon ng tiwala ay may makabuluhang prediksyon sa IU, habang ang ET, DT, HBT, at IT ay may makabuluhang prediksyon sa CE. Ang IU ay makabuluhang nakapaghula ng CE at bahagyang namamagitan sa relasyon ng DT at CE. Isinagawa ang *Template Thematic Analysis* upang tukuyin ang mga mekanismong disenyo na tumutugon sa partikular na dimensyon ng tiwala. Sa ikalawang yugto, isinagawa ang isang *design validation study* kung saan ang mga napatunayang dimensyon ng tiwala mula sa Phase 1 ay isinalin sa mga rekomendasyon sa UI design sa pamamagitan ng *triangulated design process*, at sinuri ang kanilang bisa sa pamamagitan ng kontroladong eksperimento gamit ang *Welch's t-test*. Matagumpay na nasaklaw ng ikalawang yugto ang limang dimensyon ng tiwala (CT, ET, ST, HBT, at IT), IU, at CE. Itinatag ng pananaliksik na ito ang tiwala bilang isang *multidimensional* at dinamikong nagkakalibreng konstrukto, na nag-aambag sa teorya ng *human-AI interaction* habang nagbibigay ng mga gabay sa disenyo para sa LLM

interfaces na sumusuporta sa angkop na kalibrasyon ng tiwala sa parehong yugto ng pag-aampon at patuloy na paggamit.

Keywords: Large Language Models, multidimensional trust, human-AI interaction, structural equation modeling, trust calibration, technology adoption